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Roll No

EX-6004-CBGS

B.E. VI Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Electronic Instrumentation

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Draw block diagram of CRO and write different parts with their functions.
b) With the help of suitable diagrams differentiate dual trace and dual beam CROs.
2. a) How does the digital storage oscilloscope differ from the conventional storage oscilloscope using a storage CRO? What are the advantages of each?
b) Define deflection sensitivity and deflection factor for a CRO and prove the formula for deflection sensitivity.
3. a) Derive the general equation for balance for an a.c bridge. Prove that two conditions i.e. for magnitude and phase have to be satisfied if an a.c bridge is to be balanced unlike a d.c. bridge where in only the magnitude condition is to be satisfied.
b) Describe the working of a low voltage schering bridge. Derive the equation for capacitance and dissipation factor. Draw the phasor diagram of the bridge under condition of balance.

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4.
 - a) Explain with a diagram how Maxwell's bridge can be used to measure unknown inductance.
 - b) Explain with a diagram the operation of a De-Sauty bridge.
5.
 - a) What do you understand by Electrical Transducers? State the advantages of Electrical Transducer.
 - b) Describe with the help of a diagram and characteristic the operation of LVDT.
6.
 - a) What is the difference between Analog and Digital voltmeter? Explain working principle of digital voltmeter with proper diagram.
 - b) Describe digital frequency meter and LCD with suitable diagram.
7.
 - a) Draw the block diagram of successive approximation digital voltmeter and explain its working.
 - b) Explain the IEEE 488 instrumentation bus with the help of its schematic representation.
8. Write short notes on any two of the following
 - i) Strain gauge and gauge factor
 - ii) Opto-electronic transducers
 - iii) Wave analyzer
 - iv) Spectrum analyzer
 - v) Digital Multimeter
 - vi) LED and LCD
